AMD Unsloth GRPO Fine-tuning

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Notebook access

notebooks.amd.com/ind-dev

Password: RadeonUnslothTuning

Public

Agenda

01 **Fine-tuning: what and when**
The problem it solves

02 **Types of fine-tuning**
LoRA, QLoRA, SFT, GRPO

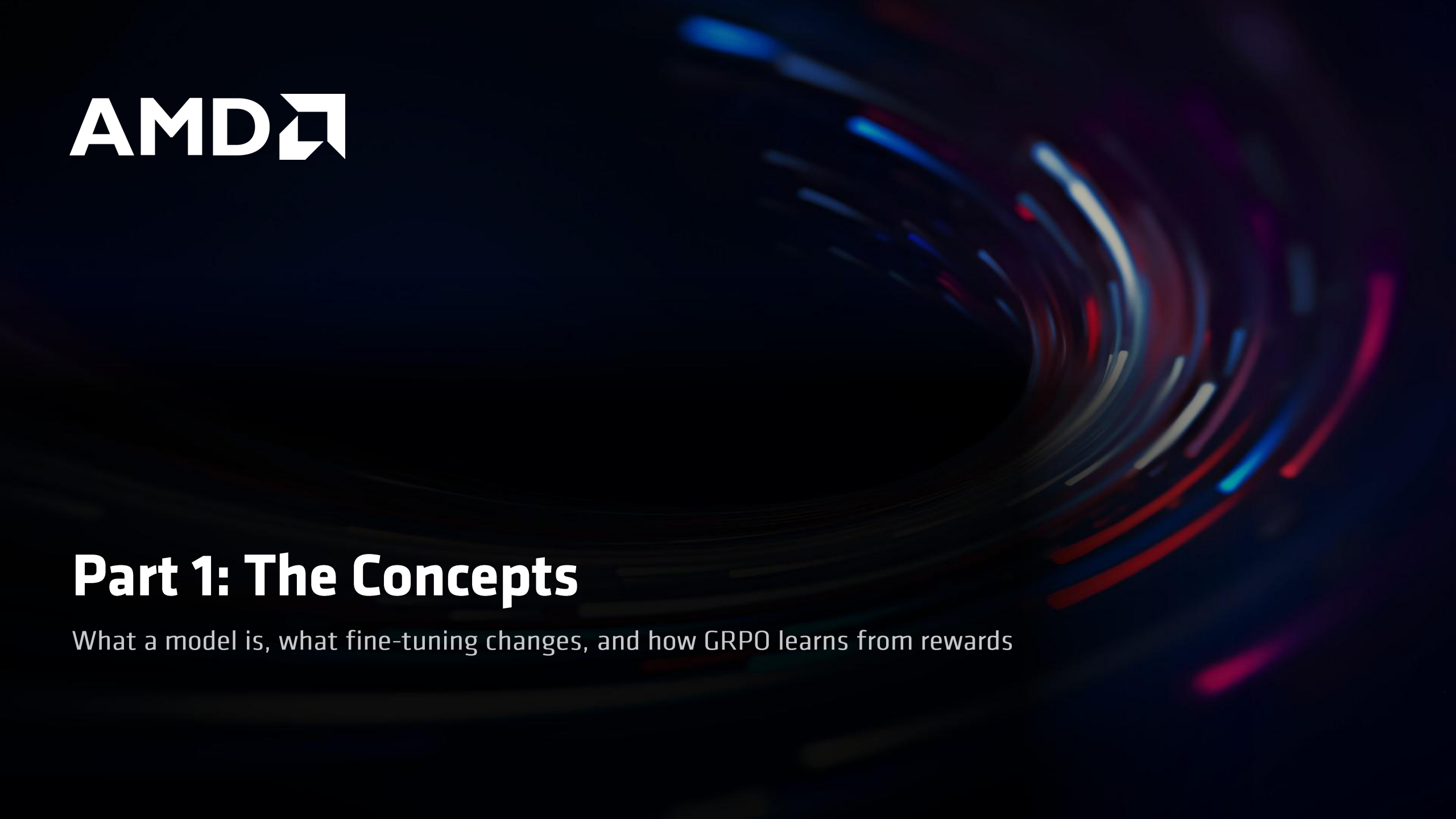
03 **The Unsloth advantage**
Faster and leaner on one GPU

04 **A game of Sudoku**
Our reward-driven task

05 **Radeon PRO W7900D**
rocm-smi and rocminfo

06 **Hands-on notebook**
Run it yourself

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Part 1: The Concepts

What a model is, what fine-tuning changes, and how GRPO learns from rewards

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What is a Large Language Model?

A neural network trained on a huge amount of text so it can predict **the next token** given what came before.

This ability enables models to chat, summarize, write code and reason.



Base models

Raw next-token predictors. Great starting point for training but awkward to chat with.

Instruct models

Further trained to follow instructions and hold a conversation.

Generalists

Llama, Qwen, Mistral on Hugging Face know a bit of everything, but not something specific to you.

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What is fine-tuning? Why do it?

Take a pre-trained model and keep training it a little more on **your** data, so it adapts to **your** task, domain, tone or format.

1

Voice

Speak in a specific style or persona: your brand, your support agent.

2

Domain

Know legal, medical, or your own product documentation.

3

Format

Always return JSON. Always cite sources.

4

Focus

Do one narrow task well on a small, cheap model.

Often cheaper and more reliable than a very long prompt or a bigger model. **A small model can be enough.**

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LoRA and QLoRA

Parameter-efficient fine-tuning: freeze the model, train a small adapter.

LoRA Low-Rank Adaptation

- Inject small low-rank matrices into chosen layers
- Original weights stay frozen, only adapters are trained
- Trainable parameters drop by orders of magnitude

QLoRA Quantized Low-Rank Adaptation

- Everything LoRA does, plus a 4-bit base model
- Cuts VRAM again, so bigger models fit on one GPU
- The usual answer to an out-of-memory error

1.16%

of Gemma 4 E2B trained today

4-bit

base weights under QLoRA

Comparable

quality to full fine-tuning

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Training methodology: SFT and GRPO

Two ways to tell a model it did well. SFT shows the answer, while GRPO scores attempts.

SFT Supervised Fine-Tuning

- Uses labelled data: prompt in, correct answer out
- The model imitates the answers you supply

```
{"instruction": "Summarize in one line.",
  "input":  "<article text>",
  "output": "<summary>"}
```

GRPO Group Relative Policy Optimization

- No labelled answer needed, only a way to score one
- Model generates a group of attempts and compares them

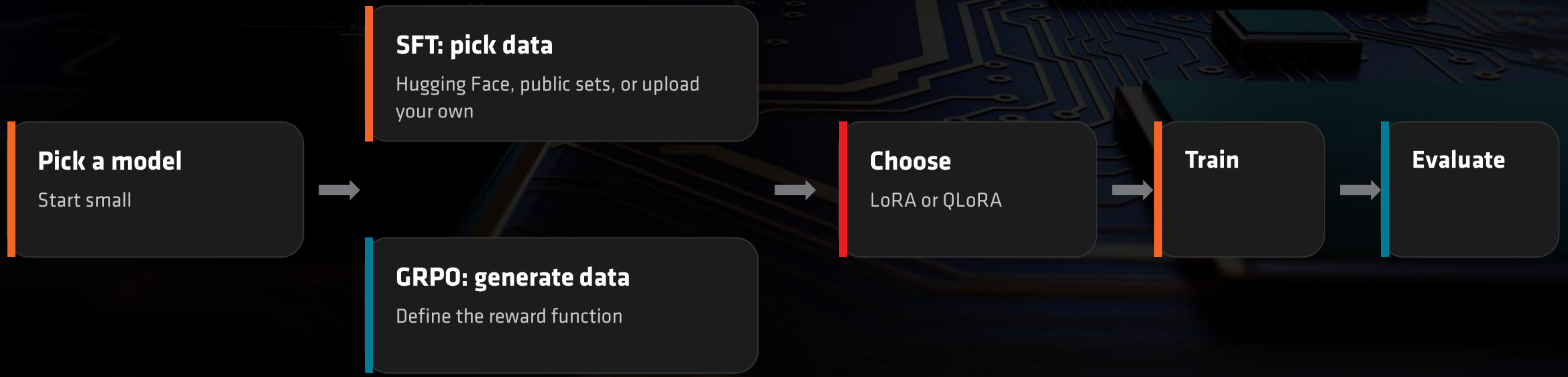
1. **Generate** several candidate answers
2. **Score** each with a reward function
3. **Prefer** the ones that scored above the group

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The fine-tuning pipeline

The steps are the same for both. Only the data step changes.



Training knobs

`learning rate` `number of steps` `time limit` `max sequence length`

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Unsloth

An open-source library for training, running and exporting LLMs **locally**, with less memory and in less time.

It builds on Hugging Face Transformers and TRL, so code you already have still works.

LoRA and QLoRA

built-in

Quantization

4-bit and 8-bit paths

Optimized kernels

faster attention and MLP

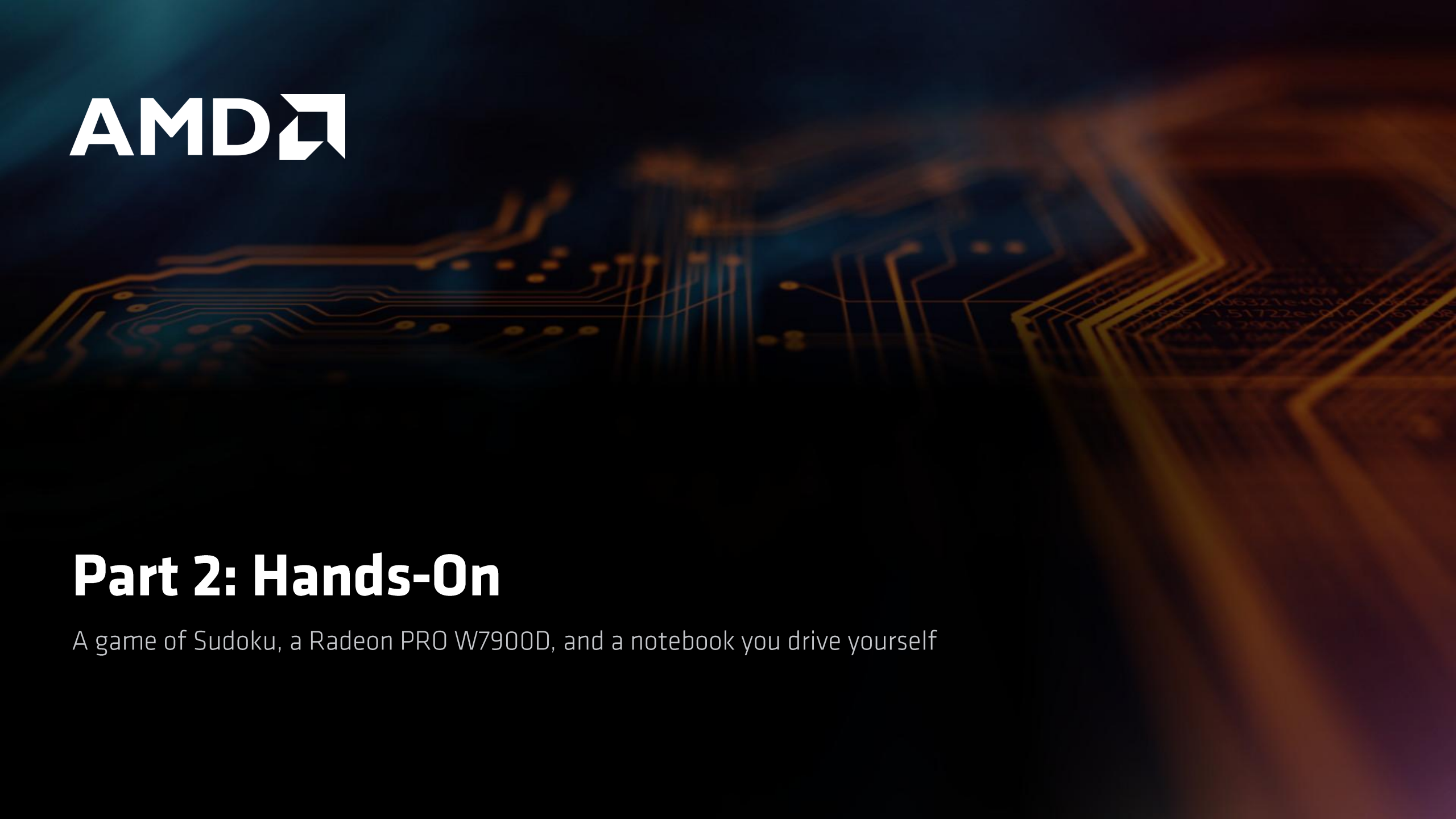
Consumer hardware

one GPU is enough

In this workshop

Unsloth fits fifteen GRPO steps into about 12½ minutes on one Radeon PRO W7900D.

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Part 2: Hands-On

A game of Sudoku, a Radeon PRO W7900D, and a notebook you drive yourself

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A game of Sudoku

Sudoku means **“single number”** in Japanese.

A logic puzzle: place digits 1 to 9 so every row, every column and every 3x3 box contains each digit exactly once.

Verifiable

A move is legal or it isn't. No human judge needed.

Partial credit

Points for code that runs, for each legal move, and most for a solve.

Hard to cheat

Imports are blocked and every run has a time limit.

PROMPT

Write strategy(board, initial) that returns the next move. Plain Python, no imports.

↓ Gemma 4 E2B writes

STRATEGY (excerpt)

```
def strategy(board, initial):
  for r in range(9):
    for c in range(9):
      if board[r][c] == 0:
        for num in range(1, 10):
          if is_valid(board, r, c, num):
            return (r, c, num)
```

REWARD, ON A PUZZLE WITH 40 BLANKS

Code runs	+1
No imports	+1
Each legal move	+0.2
Puzzle solved	+30

The model writes a strategy function. The notebook plays it and scores it.

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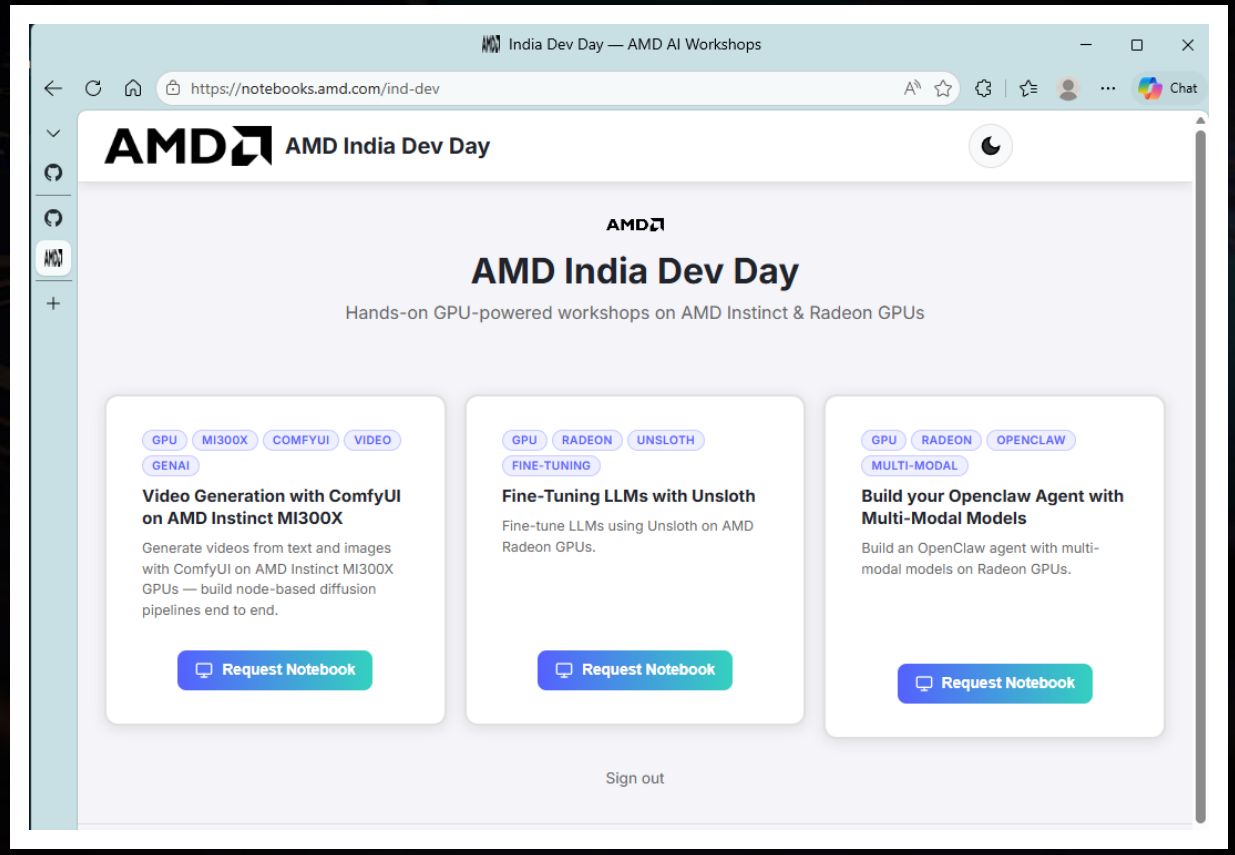
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AMD Unsloth fine-tuning workshop

1. **Open the portal**
   `notebooks.amd.com/ind-dev`

2. **Enter the password**
   `RadeonUnslothTuning`

3. **Pick the Unsloth card**
   Lands you on a Radeon PRO W7900D with 48 GB



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GPU visibility and health

Before you train, open a terminal in Jupyter and check that the GPU is visible.

terminal

```
$ rocm-smi
$ rocminfo
```

rocm-smi

Live view: temperature, power, clocks, VRAM use

rocminfo

Static view: agents, ROCm version, GPU identity

GPU% stuck at 0 while a cell runs? It is not training.

```
$ rocm-smi
```

Device	Temp	Power	SCLK	MCLK	Fan	PwrCap	VRAM%	GPU%
0	27.0°C	7.0W	0Mhz	96Mhz	20.0%	241.0W	0%	0%

```
$ rocminfo
Runtime Version:    1.18
...
Name:               gfx1100
Device Type:        GPU
```

rocminfo lists the CPU as an agent too. The Radeon PRO W7900D is the agent named gfx1100.

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Driving the notebook

Run cells top to bottom. Loading the model and the first generation are the slow parts.

Watch the marker

`In [*]` means still running, `In [7]` means done

Do not skip ahead

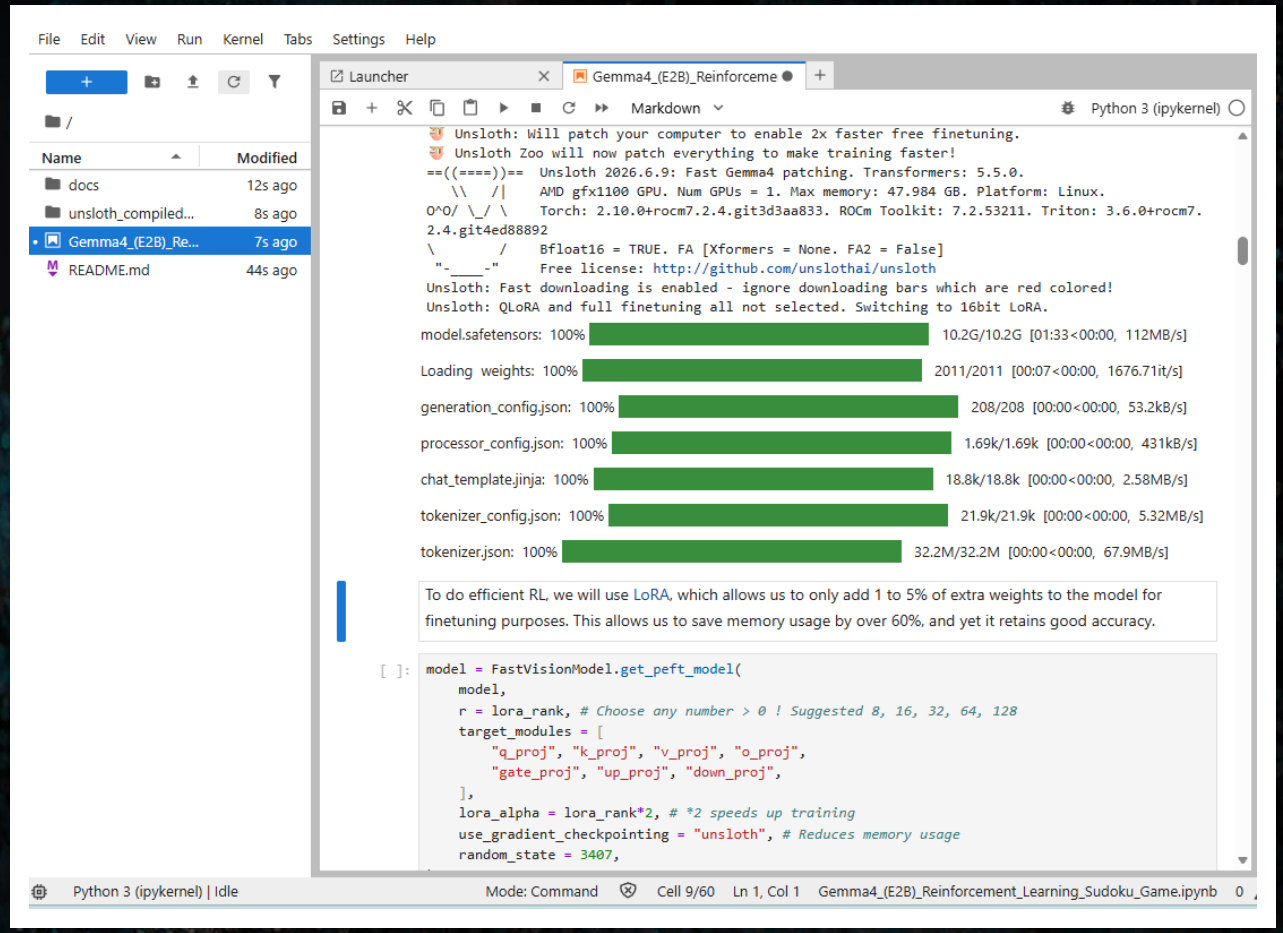
Later cells depend on variables the earlier ones define

Keep an eye on the GPU

Second terminal tab running `rocm-smi`

Cold start is normal

The warmup cell takes the first-generation cost before training starts.



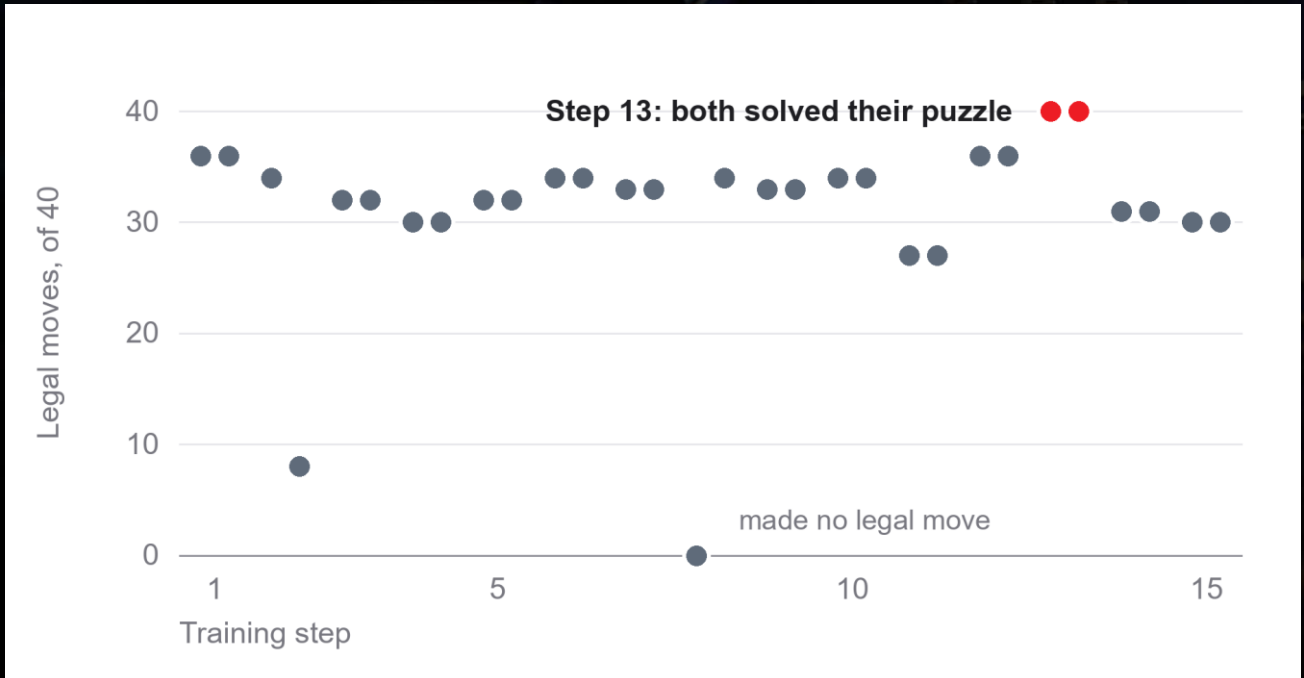
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Fine-tuning complete

Fifteen GRPO steps on the portal's Radeon PRO W7900D took about 12½ minutes.

DEMO: 15 STEPS



Legal moves, out of 40, for each strategy the model wrote (two per step)

3	5	8	1	2	4	7	6	9
1	2	7	6	9	3	4	5	8
6	9	4	7	5	8	1	3	2
5	3	6	8	1	2	9	4	7
7	8	1	5	4	9	3	2	6
2	4	9	3	6	7	8	1	5
4	1	2	9	8	6	5	7	3
8	7	5	2	3	1	6	9	4
9	6	3	4	7	5	2	8	1

STEP 13 STRATEGY, NEW PUZZLE

40 of 40 moves

Red digits are its moves. It solved 1 of 20 new puzzles like this one.

The step 13 strategy replayed on a new puzzle

What to look for

- Most strategies make 27 to 36 legal moves, then guess wrong
- It never backtracks, so a solve is partly luck. Longer runs do better

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Troubleshooting

SYMPTOM	FIX
First generation is slow	Kernels compile on first use. Run the warmup cell; later calls are faster.
Reward stays negative	The model isn't returning a Python code block. Re-run the prompt cell unchanged.
Out of memory during training	Lower max_seq_length or lora_rank. num_generations is already at its minimum of 2.
Training loss stays near 0	Normal for GRPO. Watch the reward column instead.
Cell never completes	Run rocm-smi. If GPU% stays at 0, restart the kernel and run from the top.
What to try next	Raise max_steps, or resume from outputs/checkpoint-15 as the notebook shows.

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Learn more

Unsloth	github.com/unslothai/unsloth	The framework itself
Unsloth docs	docs.unsloth.ai	Guides, notebooks, supported models
LoRA paper	arxiv.org/abs/2106.09685	Low-Rank Adaptation, Hu et al.
Gemma 4 E2B	huggingface.co/unsloth/gemma-4-E2B-it	The model used today
GRPO / DeepSeekMath	arxiv.org/abs/2402.03300	Where GRPO is introduced
Workshop notebook	notebooks.amd.com/ind-dev	Password: RadeonUnslothTuning

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